

Math Pattern Detective

Grade 1 Lesson Plan

PLANRS ACADEMY

Innovative learning for school students with
fun, culturally relevant math for Grade 1

Let's explore patterns all around us!

What Is a Pattern?

A pattern is something that **repeats in a predictable order**. Look for things that come again and again!

Patterns help us **predict what comes next**. Like:

Red – Blue – Red – Blue – ???

Answer: Red! The pattern is Red–Blue repeating.

Patterns Around Us!

We see patterns everywhere in our daily lives:

Saree Borders	Repeating geometric motifs along fabric edges
Rangoli Art	Symmetrical colour and shape designs at doorsteps
Tabla Beats	Repeating rhythmic sound sequences (Dha-Dhin-Dhin-Dha)
Kolam Designs	Dot-and-loop patterns drawn on floors
Mehndi	Repeating floral and paisley designs on hands
Flowers	Repeating petals around a centre
Leaves	Alternating leaf shapes on a stem
Floor Tiles	Square/hexagonal tiles repeating across a surface

The Repeating Unit (Core Unit)

Every pattern has a **core unit** – the smallest piece that keeps repeating.

AB Pattern

Red bead, Blue bead, Red bead, Blue bead ...

Core = Red – Blue

ABC Pattern

Mango, Banana, Guava, Mango, Banana, Guava ...

Core = Mango – Banana – Guava

AAB Pattern

Clap, Clap, Stomp, Clap, Clap, Stomp ...

Core = Clap – Clap – Stomp

Types of Patterns

1. Colour and Shape Patterns

Patterns use **colours** and **shapes** that repeat! Look for Red–Blue patterns in Holi celebrations and Circle–Square patterns in bindis and kites.

Colour Patterns	Shape Patterns	Cultural Motifs
Red – Blue – Red – Blue	Circle – Square – Circle	Bindis, Kites, Holi colours!

2. Size and Sound Patterns

Patterns aren't just what we **see** – they're also what we **hear** and **feel**!

Size Patterns	Sound Patterns
Big – Small – Big – Small ...	Clap – Stomp – Clap – Stomp ...
Try with your hands: stretch wide, then small, repeat!	Action games make learning fun – try in class!

3. Number Patterns

Counting in 2s	Pairs of students walking: 2, 4, 6, 8 ... Count your steps in twos!
Counting in 5s	5-rupee coins in a row: 5, 10, 15, 20 ... Save your coins and count!
Repeating Numbers	Roll-call fun: 1, 2, 3, 1, 2, 3 ... The numbers keep coming back!

Growing vs Repeating Patterns

Growing Patterns	Repeating Patterns
Get bigger each time	Stay the same
1, 2, 3, 4, 5 ...	1, 2, 1, 2, 1, 2 ...
<i>Example: stacking one more block each time</i>	<i>Example: beads alternating two colours</i>

Session Outline: 30 Minutes of Fun Learning

Phase	Activity	Duration
Hook & Warm-Up	Shared Practice: introduce what a pattern is	0 – 3 min
Guided Practice	What is a Pattern? – class discussion with examples	3 – 8 min
Independent Practice	Students identify and extend patterns on their own	8 – 13 min
Creative Challenge	Students design their own original pattern	13 – 22 min
Recap	Review key vocabulary and concepts together	22 – 27 min
Exit Ticket	Quick 5-question quiz to check understanding	27 – 30 min

Learning Objectives Mapped to Bloom's Taxonomy

Our young detectives climb Bloom's pyramid! From recognising repeating units to explaining what repeats and why, then applying skills to extend patterns and creating their own unique designs.

Remember	Recognise AB, ABC, AAB pattern types	Recall the word <i>pattern</i> and give an example
Understand	Explain why something is a pattern	Describe the repeating unit in own words
Apply & Analyse	Add elements to extend a given pattern	Create an original 2-element repeating pattern

Micro-Learning Objectives

LO 1–2	Know & Identify Patterns Say the word " pattern " and give an example from your life. Spot the repeating unit in colour or shape patterns!
LO 3–4	Extend & Recognise Extend an AB pattern by 2 or more elements . Identify ABC or AAB patterns like Mango–Banana–Guava!
LO 5	Create Your Own Pattern Design and build your very own 2-element repeating pattern independently. You're a Pattern Detective now!

Success Criteria for Our Young Detectives

Our young pattern detectives will succeed when they can:

Define It	Explain what a pattern is in their own words
Spot It	Identify repeating units with 90% accuracy
Extend & Create	Add 2 elements to an AB pattern correctly
Score	Achieve 3 out of 5 or above on the Exit Ticket quiz
Real-World Link	Spot at least one pattern at home or in school

Celebrating Indian Culture through Patterns

This lesson connects math to the rich visual traditions of India. By exploring patterns in familiar cultural contexts, students build number sense and pattern recognition in a joyful, meaningful way.

Festive Designs	Rangoli, Kolam, Mehndi
Traditional Art	Torans, Florals, Geometric Motifs
Colour Traditions	Diyas, Festive Decor

Thank You, Pattern Detectives!

Let's discover patterns everywhere!

Visit us at planrsacademy.com

Share your patterns with us!